Name: _____

WORKSHEET

Contrapositive

1 For each these P and Q statements:

i write the original statement $P \Rightarrow Q$.

ii identify $\overline{P}$ and $\overline{Q}$.

iii write the contrapositive statement $\overline{Q} \Rightarrow \overline{P}$.

a P = you smile Q = you are happy

b P = you pat a cat Q = the cat purrs

c P = increase exercise Q = increase fitness

d P = a rhombus Q = four sides equal in length

2 Write the contrapositive of the following statements.

a If a drink is cold then it is refreshing.

b If a shape is a triangle, then it has three sides.

c All birds have feathers.

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d All linear graphs are one-to-one functions.

e If $x = 4$, then $x^2 + 1 = 17$.

f If $x < y$, then $x - y < 0$.

3 Given the contrapositive statements below, write down each original statement.

a if a shape does not have four sides, it is not a trapezium.

b if grass is not green, then it has not been raining.

c if a number is not even then it is not divisible by 2.

d if you get no correct answers on a test, you cannot pass.

e if a printer does not have ink then will not print.

f if an animal does not have a tail then it is not a possum.

4 For each of these statements,

i identify P and Q.

ii verify that $P \Rightarrow Q$ is true.

iii verify that the contrapositive $\overline{Q} \Rightarrow \overline{P}$ is also true.

a if an animal is a turtle, then it has a shell.

b If there is thunder, then there is lightening.

c if a shape is an octagon, then it has eight sides.

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d if x is positive, then x^2 is also positive.

e if a triangle has a right angle, it has a hypotenuse.

f For a positive: if $a \geq 4$, then $a^2 + a \geq 20$.

5 For each of these statements:

i write the contrapositive statement.

ii state whether the statement and its contrapositive are both true or both false.

a if you are tall you are good at basketball.

b if an animal lays eggs then it is a bird.

c if a shape is a trapezium, it has a pair of parallel sides.

d if a graph is a parabola, then it has a minimum.

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e if x is divisible by 4, then $\sqrt{x}$ is divisible by 2.

f if ab is negative then either a or b must be negative.

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Answers

1 a i If you smile then you are happy.

ii $\overline{P}$ = you don't smile $\overline{Q}$ = you are not happy

iii $\overline{Q} \Rightarrow \overline{P}$ = If you are not happy then you don't smile.

b i If you pat a cat then the cat purrs.

ii $\overline{P}$ = you don't pat a cat $\overline{Q}$ = the cat does not purr

iii $\overline{Q} \Rightarrow \overline{P}$ = If the cat does not purr you are not patting it.

c i If you increase your exercise then you increase your fitness.

ii $\overline{P}$ = you don't increase your exercise $\overline{Q}$ = your fitness does not increase

iii $\overline{Q} \Rightarrow \overline{P}$ = If your fitness does not increase then you haven't increased your exercise.

d i If a shape is a rhombus then it has four sides equal in length.

ii $\overline{P}$ = a shape is not a rhombus $\overline{Q}$ = it does not have four sides equal in length

iii $\overline{Q} \Rightarrow \overline{P}$ = If a shape does not have four sides equal in length then it is not a rhombus.

2 a If a drink is not refreshing then it is not cold.

b If a shape does not have three sides then it is not a triangle.

c If something does not have feathers, then it is not a bird.

d If a function is not one-to-one, then it cannot be a linear graph.

e If $x^2 + 1 \neq 17$, then $x \neq 4$.

f If $x - y > 0$, then $x > y$.

3 a If a shape is a trapezium, then it has four sides.

b If it has been raining, then the grass is green.

c If a number is divisible by 2, then it is even.

d If you pass a test, you must get some correct answers.

e If a printer can print, then it must have ink.

f If an animal is a possum, then it has a tail.

4 a i P = animal is a turtle, Q = it has a shell

ii Statement is true.

iii $\overline{Q} \Rightarrow \overline{P}$ = If an animal does not have a shell then it is not a turtle. This statement is true.

b i P = there is thunder, Q = there is lightening

ii Statement is true.

iii $\overline{Q} \Rightarrow \overline{P}$ = if there is not lightening then there is not thunder. This statement is true.

c i P = shape is an octagon, Q = it has eight sides

ii Statement is true.

iii $\overline{Q} \Rightarrow \overline{P}$ = if a shape does not have eight sides then it is not an octagon. This statement is true.

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d **i** $P = x$ is positive, $Q = x^2$ is positive

ii Statement is true.

iii $\overline{Q} \Rightarrow \overline{P}$ = If x^2 is not positive, then x is not positive. This statement is true.

e **i** P = a triangle has a right angle, Q = it has a hypotenuse

ii Statement is true.

iii $\overline{Q} \Rightarrow \overline{P}$ = If a triangle does not have a hypotenuse, then it does not have a right angle. This statement is true.

f **i** $P = a \geq 4$, $Q = a^2 + a \geq 20$

ii Statement is true.

iii $\overline{Q} \Rightarrow \overline{P}$ = For a positive: If $a^2 + a < 20$, then $a < 4$. This statement is true.

5 **a** **i** If you are not good at basketball then you are not tall.

ii Both statements are false.

b **i** If an animal is not a bird then it does not lay eggs.

ii Both statements are false.

c **i** If a shape does not have a pair of parallel sides, then it is not a trapezium.

ii Both statements are true.

d **i** If a graph does not have a minimum, then it is not parabola.

ii Both statements are false.

e **i** if $\sqrt{x}$ is not divisible by 2, then x is not divisible by 4.

ii Both statements are true.

f **i** if neither a nor b are negative, then ab is not negative

ii Both statements are true.

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